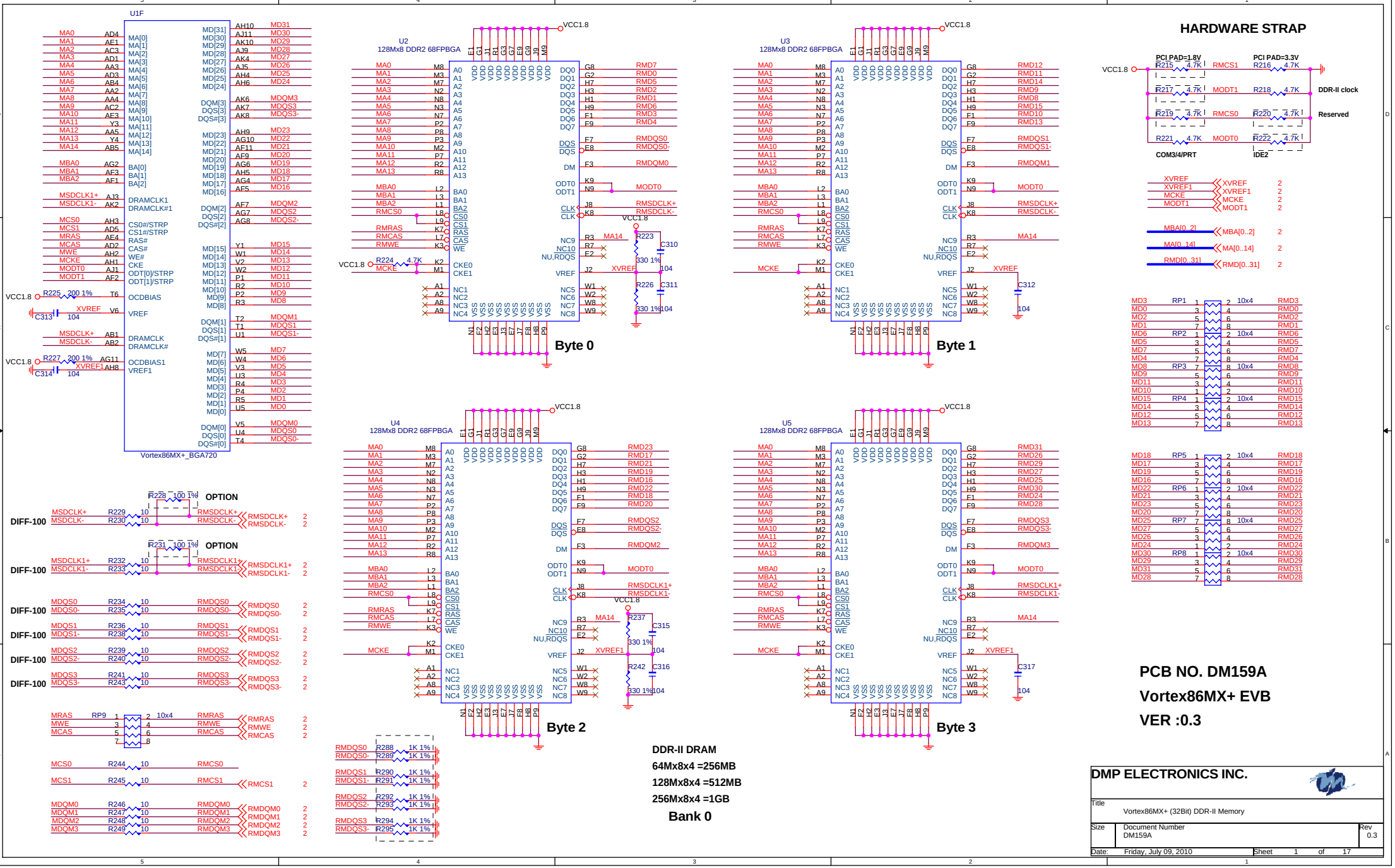
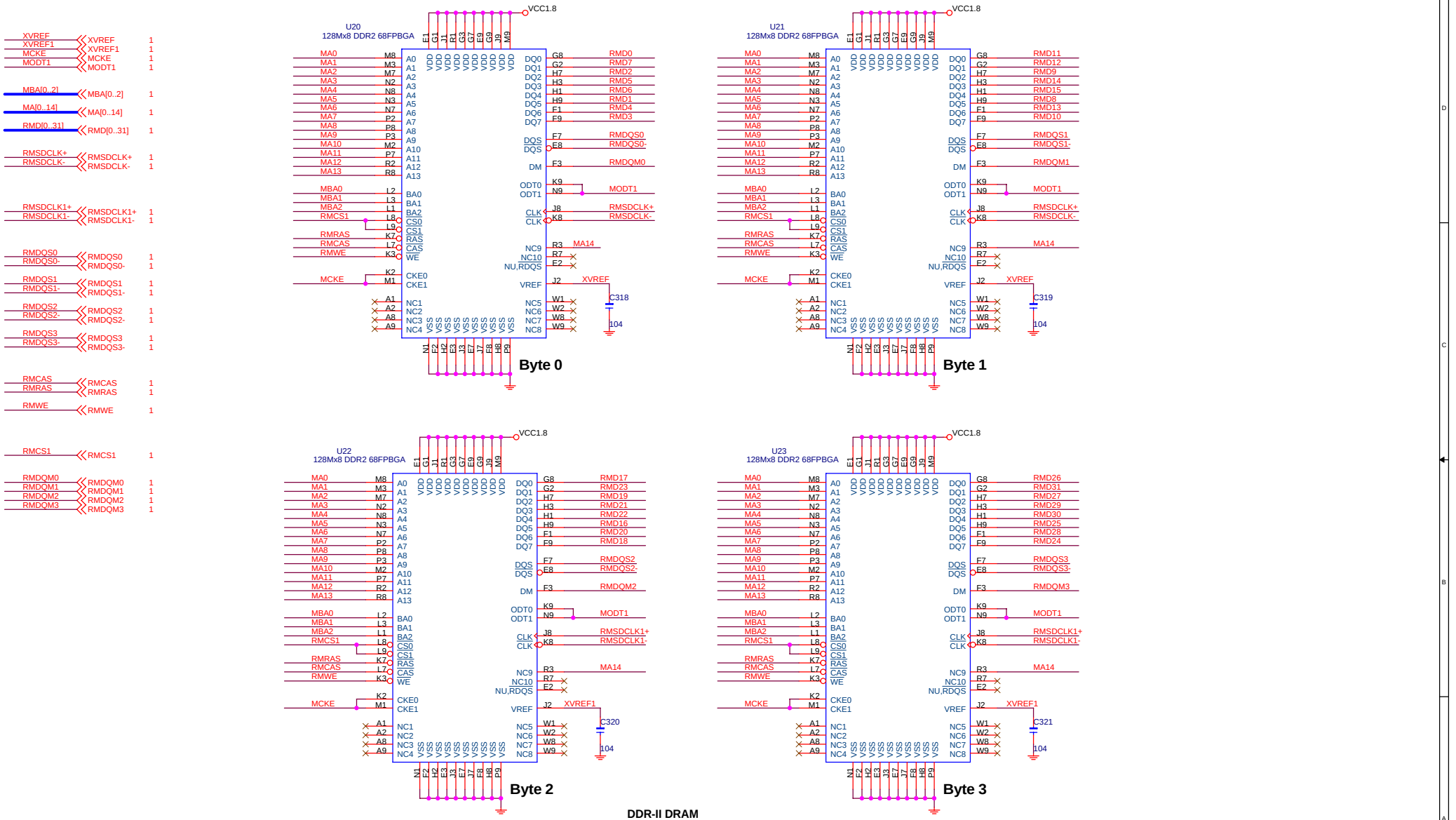


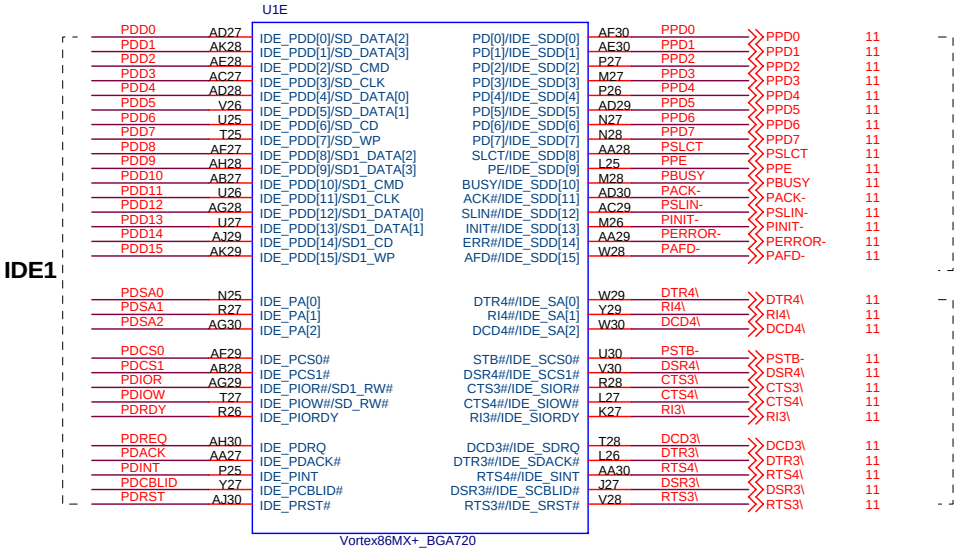
PCB NO. DM159A

Vortex86MX+ EVB

VER :0.3

DMP ELECTRONICS INC.	
Title	Vortex86MX+ (32Bit) DDR-II Memory
Size	Document Number DM159A — Rev 0.3
Date: Friday, July 09, 2010	Sheet 1 of 17


XVREF
XVREF1
MCKE
MODT1
MBA[0..2]
MA[0..14]
RMD[0..31]
RMSDCLK+
RMSDCLK-
RMSDCLK1+
RMSDCLK1-
RMDQS0
RMDQS0-
RMDQS1
RMDQS1-
RMDQS2
RMDQS2-
RMDQS3
RMDQS3-
RMCAS
RMRAS
RMWE
RMCS1
RMDQM0
RMDQM1
RMDQM2
RMDQM3
U20
128Mx8 DDR2 68FPBGA
VCC1.8
Byte 0
U21
128Mx8 DDR2 68FPBGA
VCC1.8
Byte 1
U22
128Mx8 DDR2 68FPBGA
VCC1.8
Byte 2
U23
128Mx8 DDR2 68FPBGA
VCC1.8
Byte 3
C318
C319
C320
C321
104
MODT1
MA14
XVREF
XVREF1
RMSDCLK+
RMSDCLK-
RMDQM0
RMDQM1
RMDQM2
RMDQM3
DDR-II DRAM
64Mx8x4 =256MB
128Mx8x4 =512MB
256Mx8x4 =1GB
Bank 1
DMP ELECTRONICS INC.
Title
Vortex86MX+ (32Bit) DDR-II Memory-2
Size
Document Number
DM159A
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0.3
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of
17

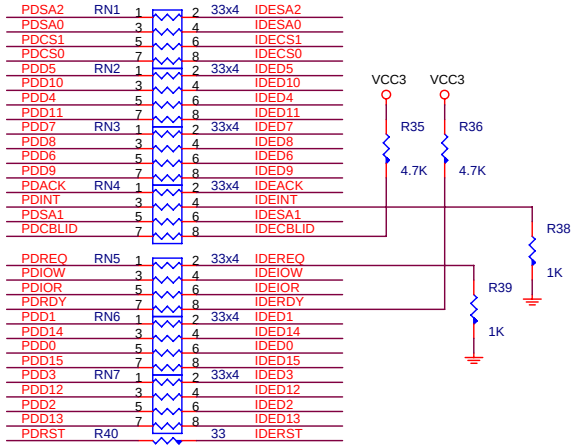


POWER LED

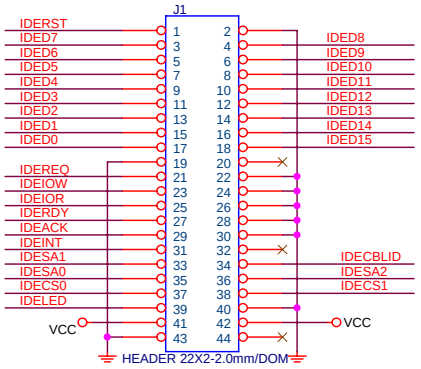


IDE LED

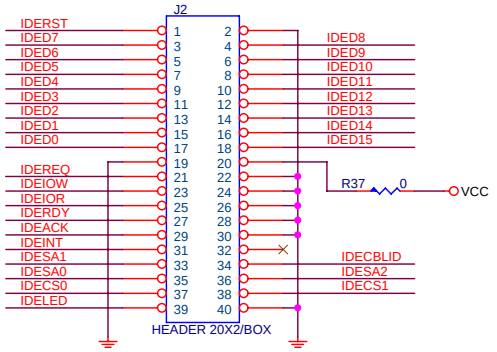




IDE 44PIN/DOM



IDE 40PIN

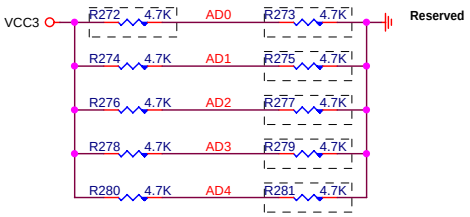


DMP ELECTRONICS INC.

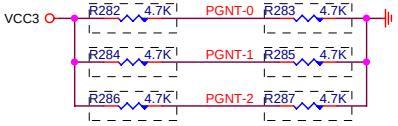
Title	Vortex86MX+ IDE	
Size	Document Number DM159A	Rev 0.3
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PCI BUS

HARDWARE STRAP



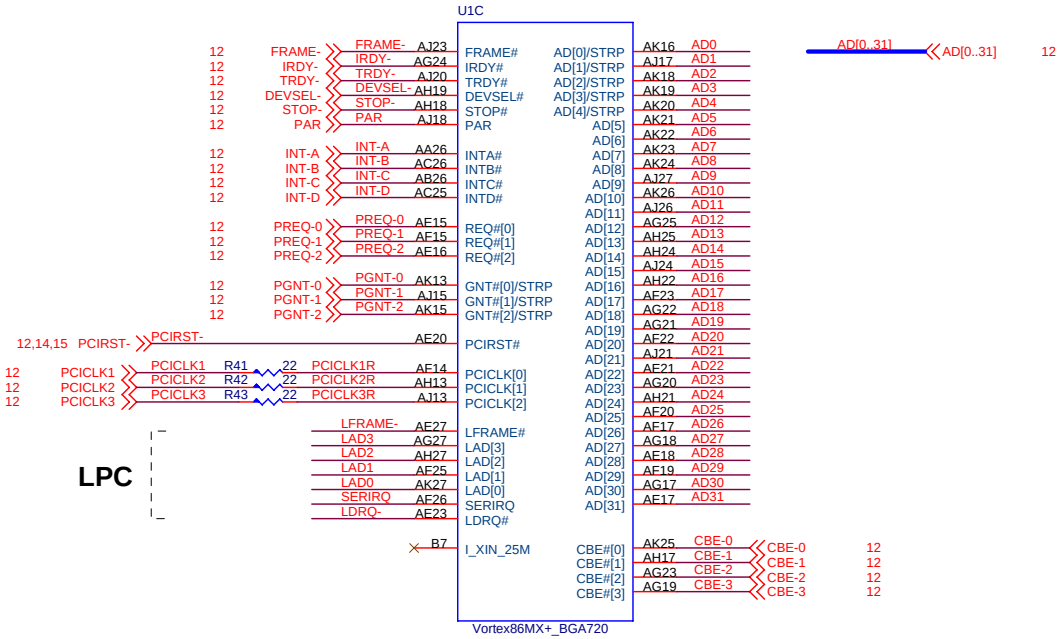
AD[4:1]---Broad ID, default 1111
(R3308+C6037, R3308+C6036)



DDRII clock select

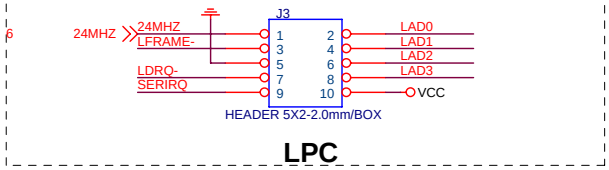
ODT1	GNT[2:0]	FREQ
0	001	200
0	010	233
0	011	266
0	100	300
0	101	333
0	110	366
0	111	400
1	000	416
1	001	433
1	010	450
1	011	466
1	100	483
1	101	500
1	110	516
1	111	533

MHz



LPC

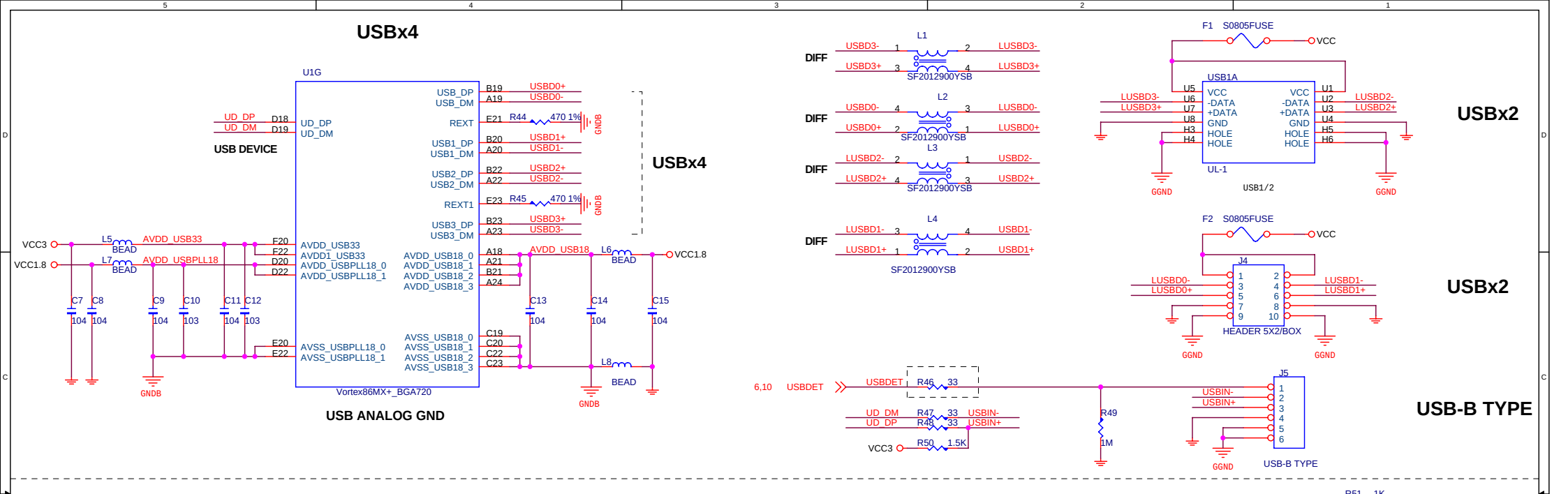
OPTION

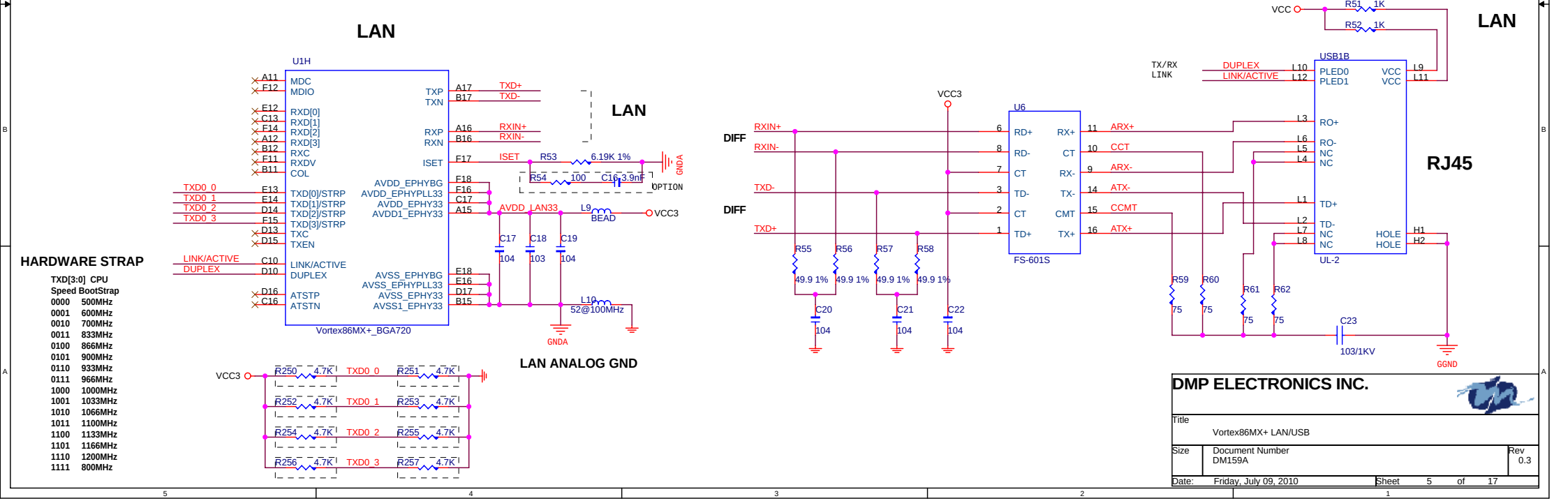


LPC

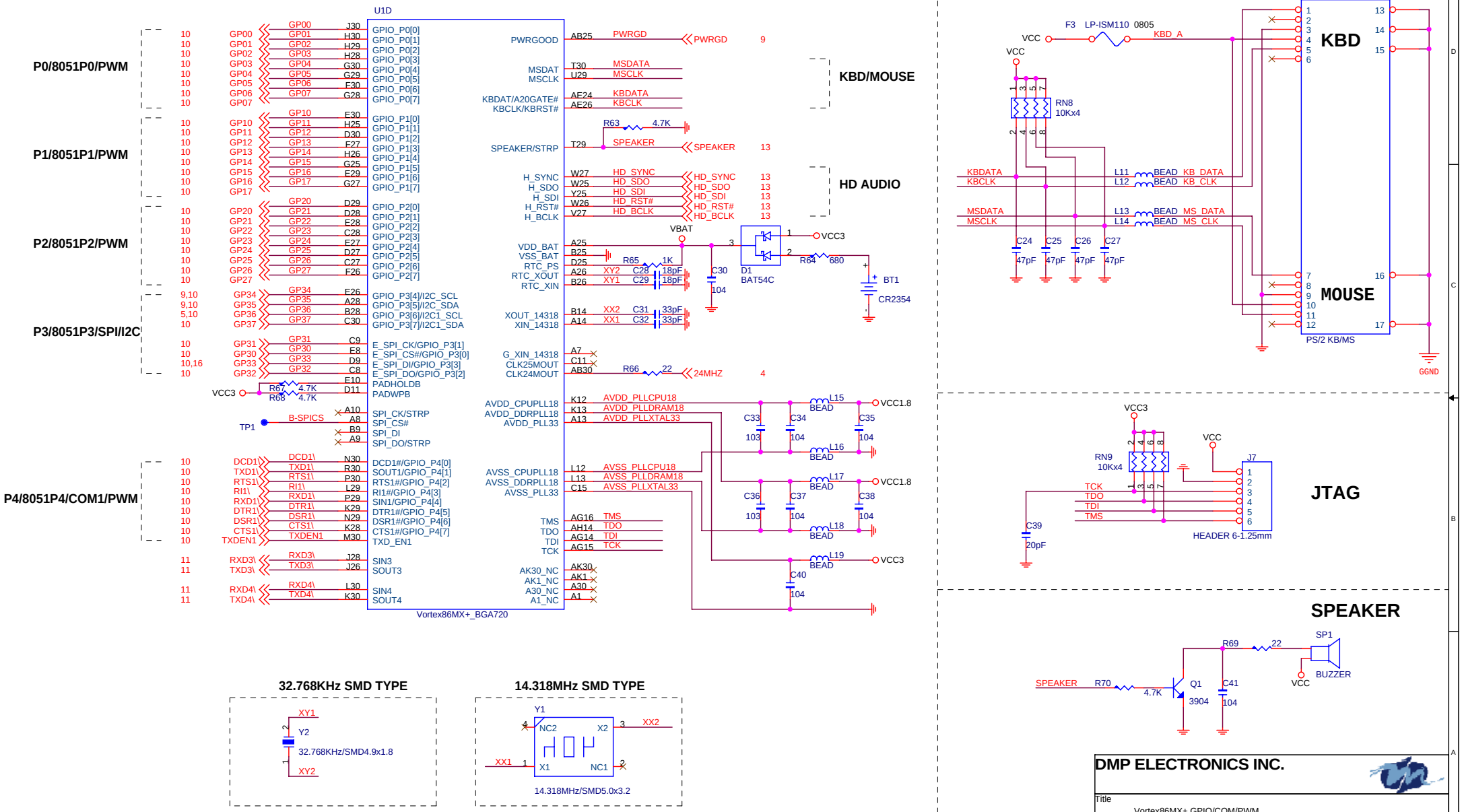
DMP ELECTRONICS INC.

Title	Vortex86MX+ PCI BUS	
Size	Document Number DM159A	Rev 0.3
Date: Friday, July 09, 2010	Sheet 4 of 17	



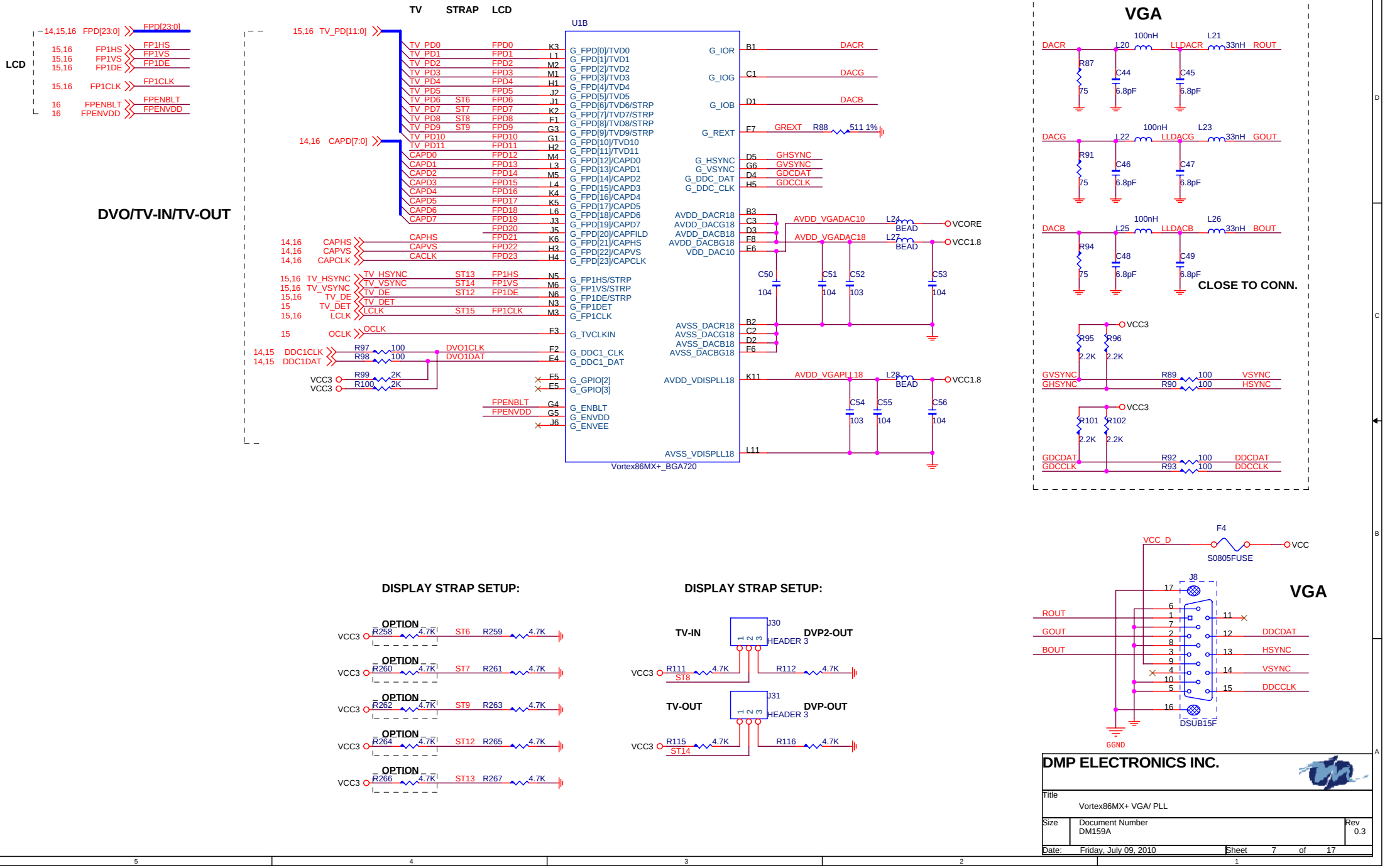


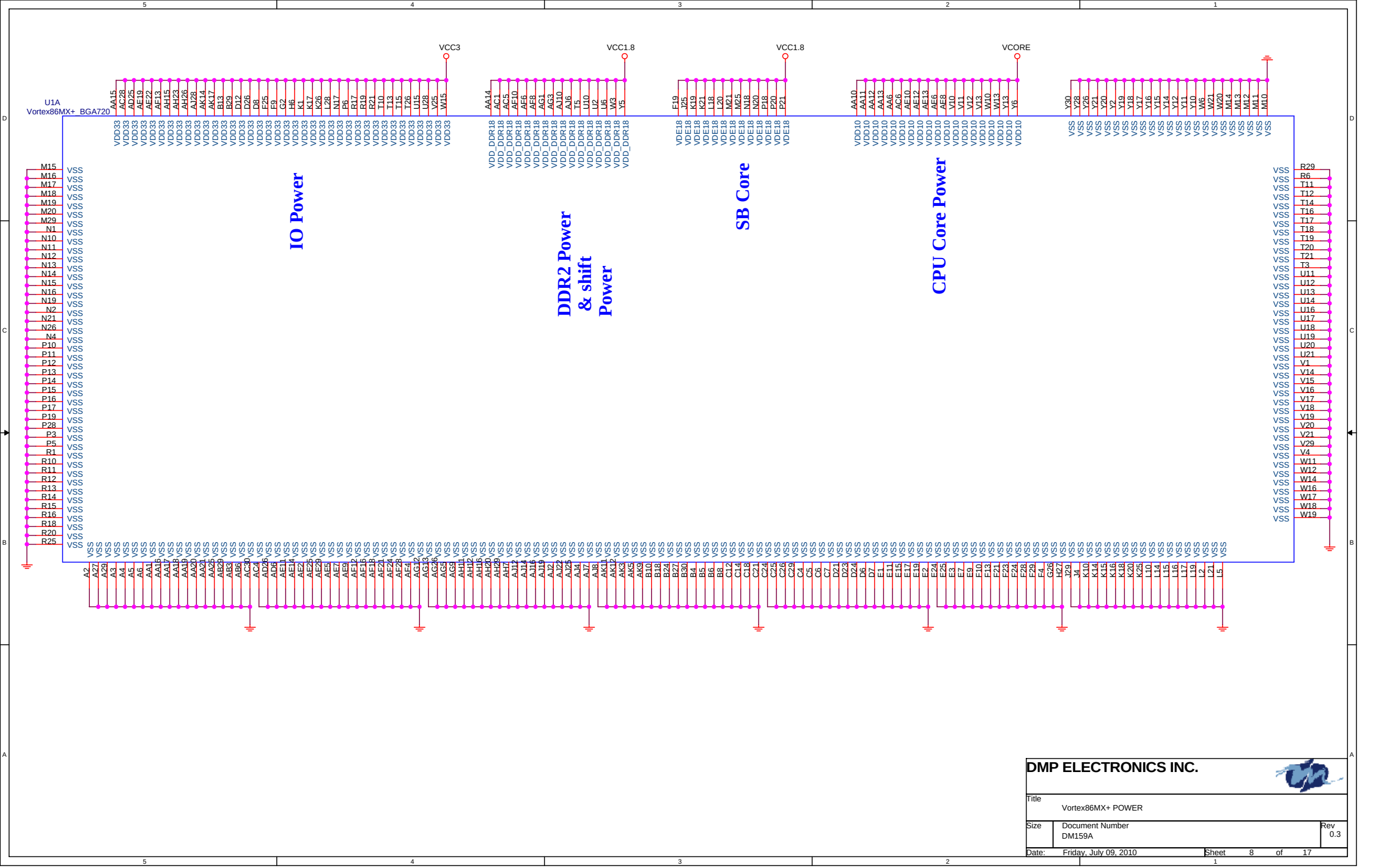
DMP ELECTRONICS INC.	
Title	Vortex86MX+ LAN/USB
Size	Document Number DM159A
Rev	0.3
Date:	Friday, July 09, 2010
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DMP ELECTRONICS INC.

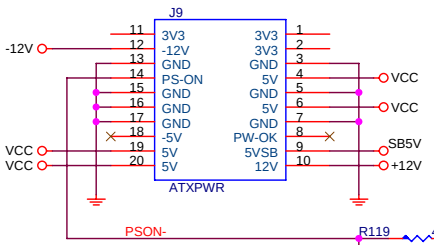
Title	Vortex86MX+ GPIO/COM/PWM	
Size	Document Number DM159A	Rev 0.3
Date: Friday, July 09, 2010	Sheet 6 of 17	

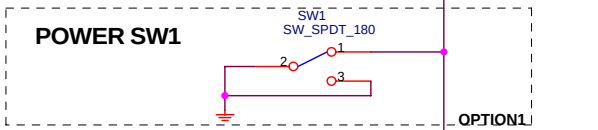

LCD
14,15,16 FPD[23:0]
FPD[23:0]
15,16 FP1HS
FP1HS
15,16 FP1VS
FP1VS
15,16 FP1DE
FP1DE
15,16 FP1CLK
FP1CLK
16 FPENBLT
FPENBLT
16 FPENVDD
FPENVDD
DVO/TV-IN/TV-OUT
TV STRAP LCD
U1B
15,16 TV_PD[11:0]
14,16 CAPD[7:0]
14,16 CAPHS
14,16 CAPVS
14,16 CAPCLK
15,16 TV_HSYNC
15,16 TV_VSYNC
15,16 TV_DE
15 TV_DET
15,16 LCLK
15 OCLK
14,15 DDC1CLK
14,15 DDC1DAT
R97 100
R98 100
R99 2K
R100 2K
VCC3
DVO1CLK
DVO1DAT
FPENBLT
FPENVDD
G_FPD[0]/TVD0
G_FPD[1]/TVD1
G_FPD[2]/TVD2
G_FPD[3]/TVD3
G_FPD[4]/TVD4
G_FPD[5]/TVD5
G_FPD[6]/TVD6/STRP
G_FPD[7]/TVD7/STRP
G_FPD[8]/TVD8/STRP
G_FPD[9]/TVD9/STRP
G_FPD[10]/TVD10
G_FPD[11]/TVD11
G_FPD[12]/CAPD0
G_FPD[13]/CAPD1
G_FPD[14]/CAPD2
G_FPD[15]/CAPD3
G_FPD[16]/CAPD4
G_FPD[17]/CAPD5
G_FPD[18]/CAPD6
G_FPD[19]/CAPD7
G_FPD[20]/CAPFILD
G_FPD[21]/CAPHS
G_FPD[22]/CAPVS
G_FPD[23]/CAPCLK
G_FP1HS/STRP
G_FP1VS/STRP
G_FP1DE/STRP
G_FP1DET
G_FP1CLK
G_TVCLKIN
G_DDC1_CLK
G_DDC1_DAT
G_GPIO[2]
G_GPIO[3]
G_ENBLT
G_ENVDD
G_ENVEE
G_IOR
G_IOG
G_IOB
G_REXT
G_HSYNC
G_VSYNC
G_DDC_DAT
G_DDC_CLK
AVDD_DACR18
AVDD_DACG18
AVDD_DACB18
AVDD_DACBG18
VDD_DAC10
AVSS_DACR18
AVSS_DACG18
AVSS_DACB18
AVSS_DACBG18
AVDD_VDISPLL18
AVSS_VDISPLL18
DACR
DACG
DACB
R88 511 1%
GREXT
GHSYNC
GVSYNC
GDCDAT
GDCCLK
AVDD_VGADAC10
AVDD_VGADAC18
AVDD_VGAPLL18
L24 BEAD
L27 BEAD
L28 BEAD
VCORE
VCC1.8
C50 104
C51 104
C52 103
C53 104
C54 103
C55 104
C56 104
Vortex86MX+_BGA720
VGA
DACR
100nH L20
LLDACR
33nH ROUT
L21
R87 75
C44 6.8pF
C45 6.8pF
DACG
100nH L22
LLDACG
33nH GOUT
L23
R91 75
C46 6.8pF
C47 6.8pF
DACB
100nH L25
LLDACB
33nH BOUT
L26
R94 75
C48 6.8pF
C49 6.8pF
CLOSE TO CONN.
VCC3
R95 2.2K
R96 2.2K
GVSYNC
GHSYNC
R89 100
R90 100
VSYNC
HSYNC
R101 2.2K
R102 2.2K
GDCDAT
GDCCLK
R92 100
R93 100
DDCDAT
DDCCLK
DISPLAY STRAP SETUP:
OPTION
R258 4.7K
ST6
R259 4.7K
R260 4.7K
ST7
R261 4.7K
R262 4.7K
ST9
R263 4.7K
R264 4.7K
ST12
R265 4.7K
R266 4.7K
ST13
R267 4.7K
DISPLAY STRAP SETUP:
TV-IN
J30
HEADER 3
DVP2-OUT
R111 4.7K
ST8
R112 4.7K
TV-OUT
J31
HEADER 3
DVP-OUT
R115 4.7K
ST14
R116 4.7K
F4
VCC_D
S0805FUSE
VCC
J8
DSUB15F
GGND
VGA
ROUT
GOUT
BOUT
DDCDAT
HSYNC
VSYNC
DDCCLK
DMP ELECTRONICS INC.
Title
Vortex86MX+ VGA/ PLL
Size
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0.3
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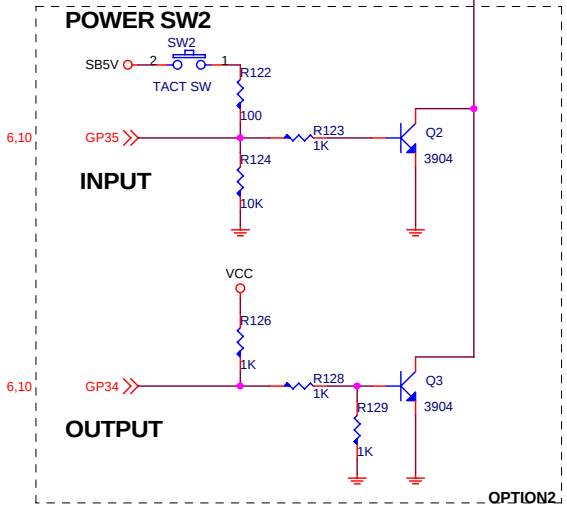


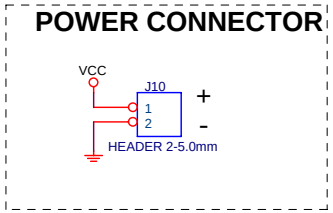
DMP ELECTRONICS INC.		
Title: Vortex86MX+ POWER		
Size	Document Number: DM159A	Rev: 0.3
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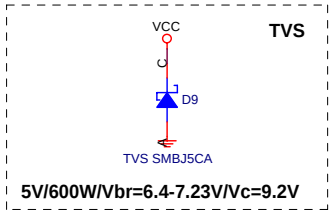
ATX POWER

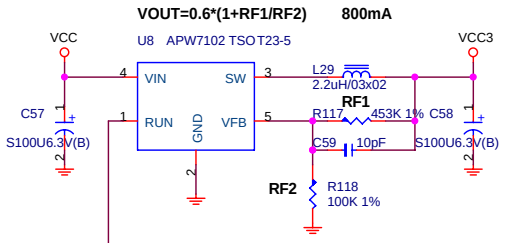


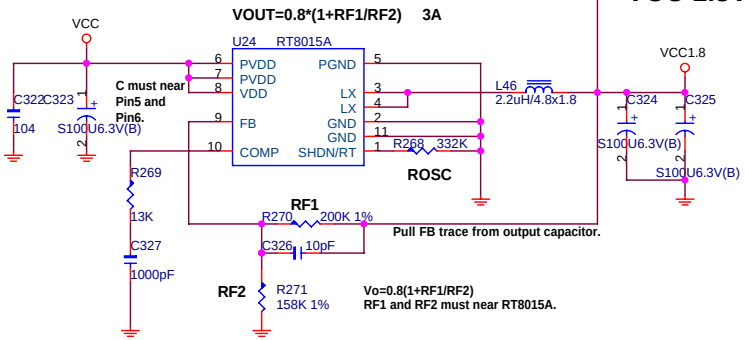


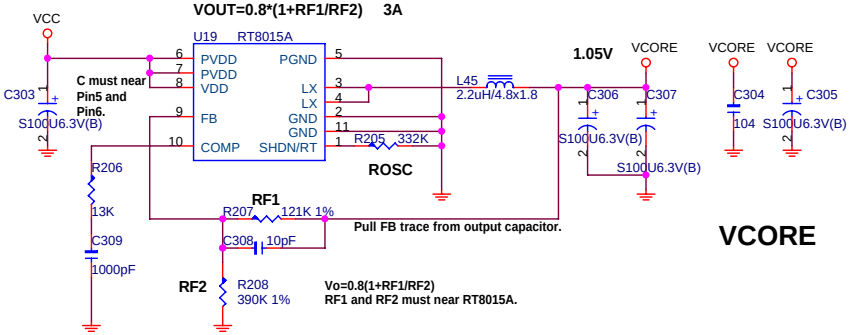


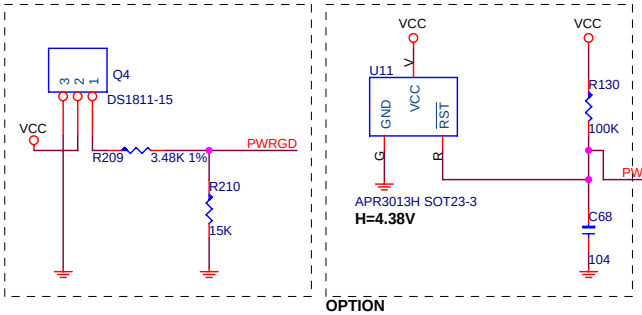




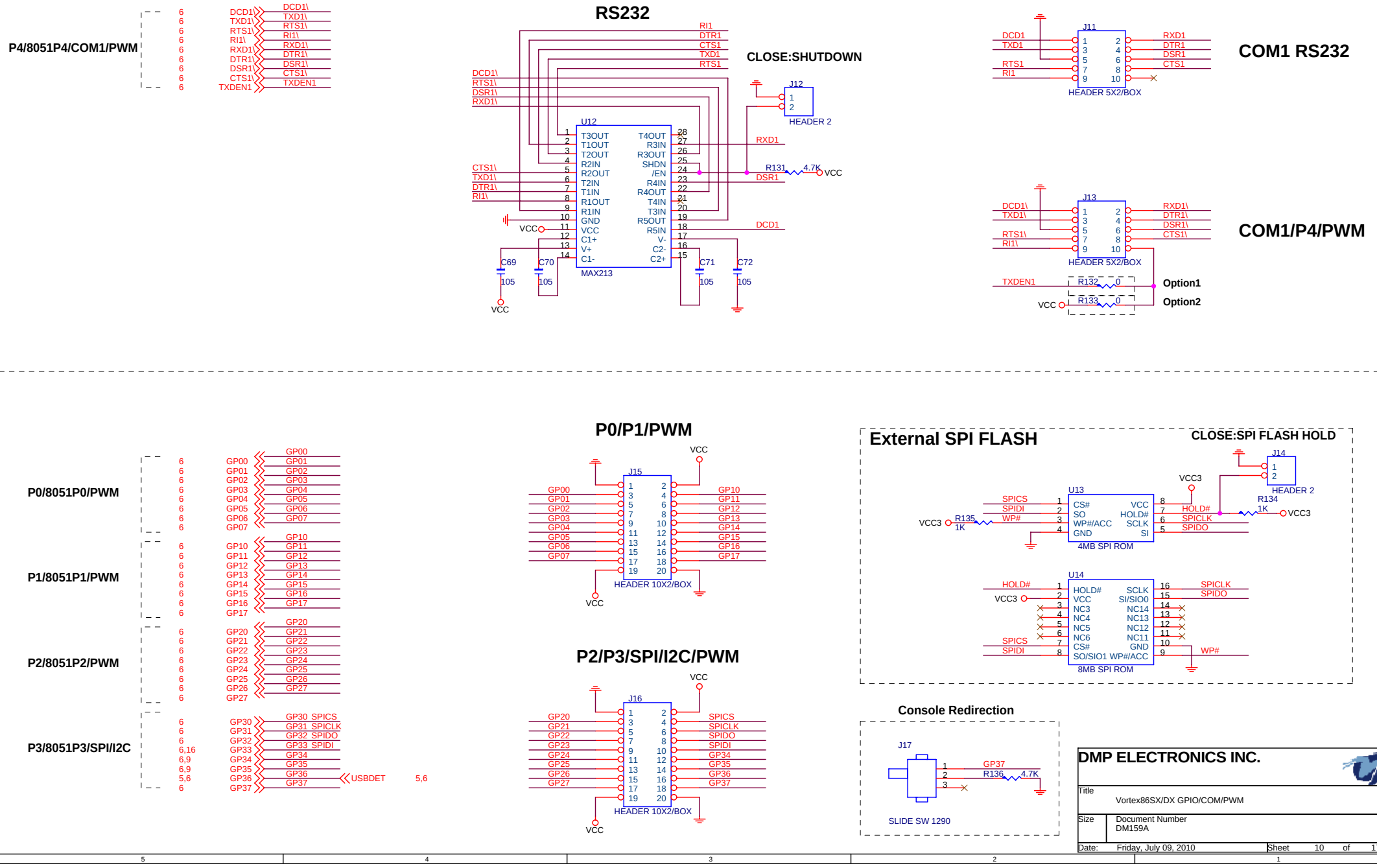








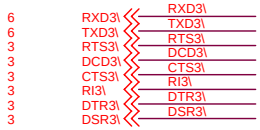
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Title: DC to DC		
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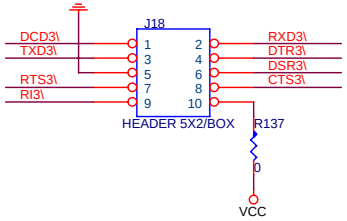


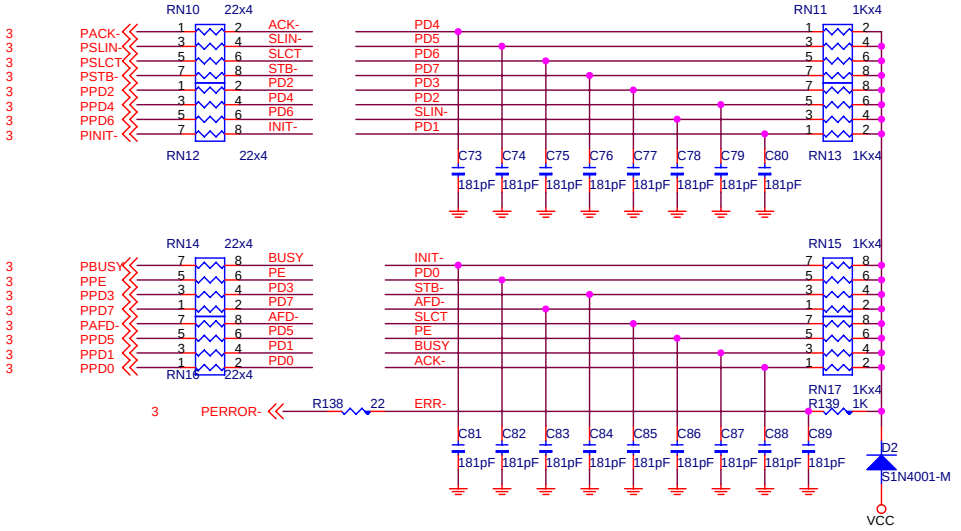
DMP ELECTRONICS INC.	
Title	Vortex86SX/DX GPIO/COM/PWM
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PRINT1/COM3/COM4

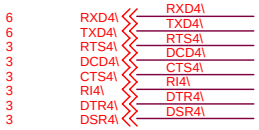
COM3 TTL

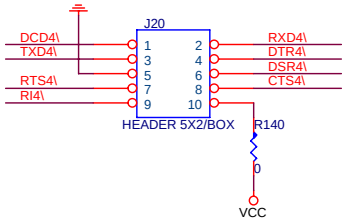


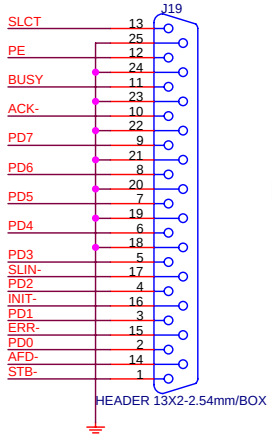




COM4 TTL



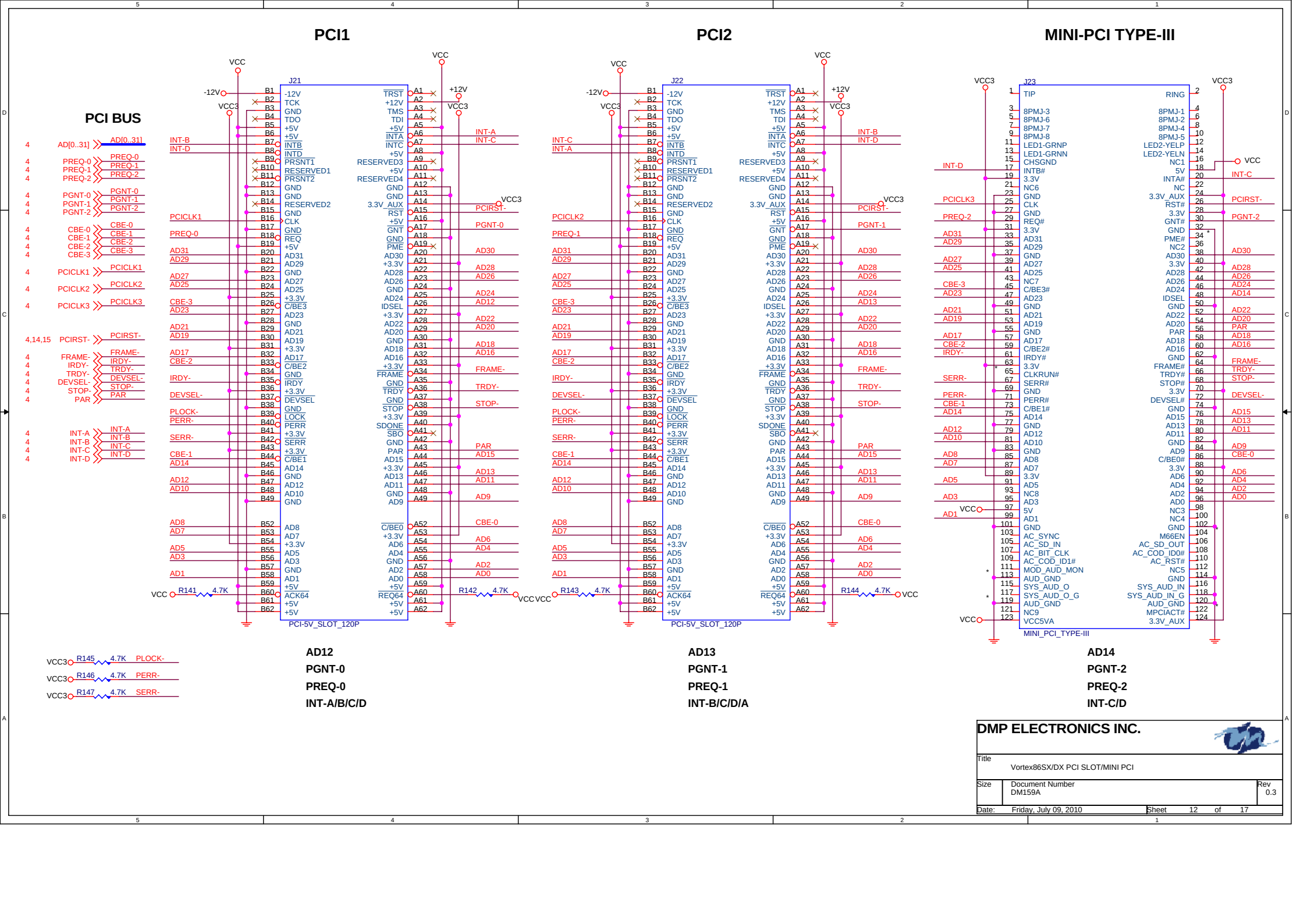




PRINT1

DMP ELECTRONICS INC.

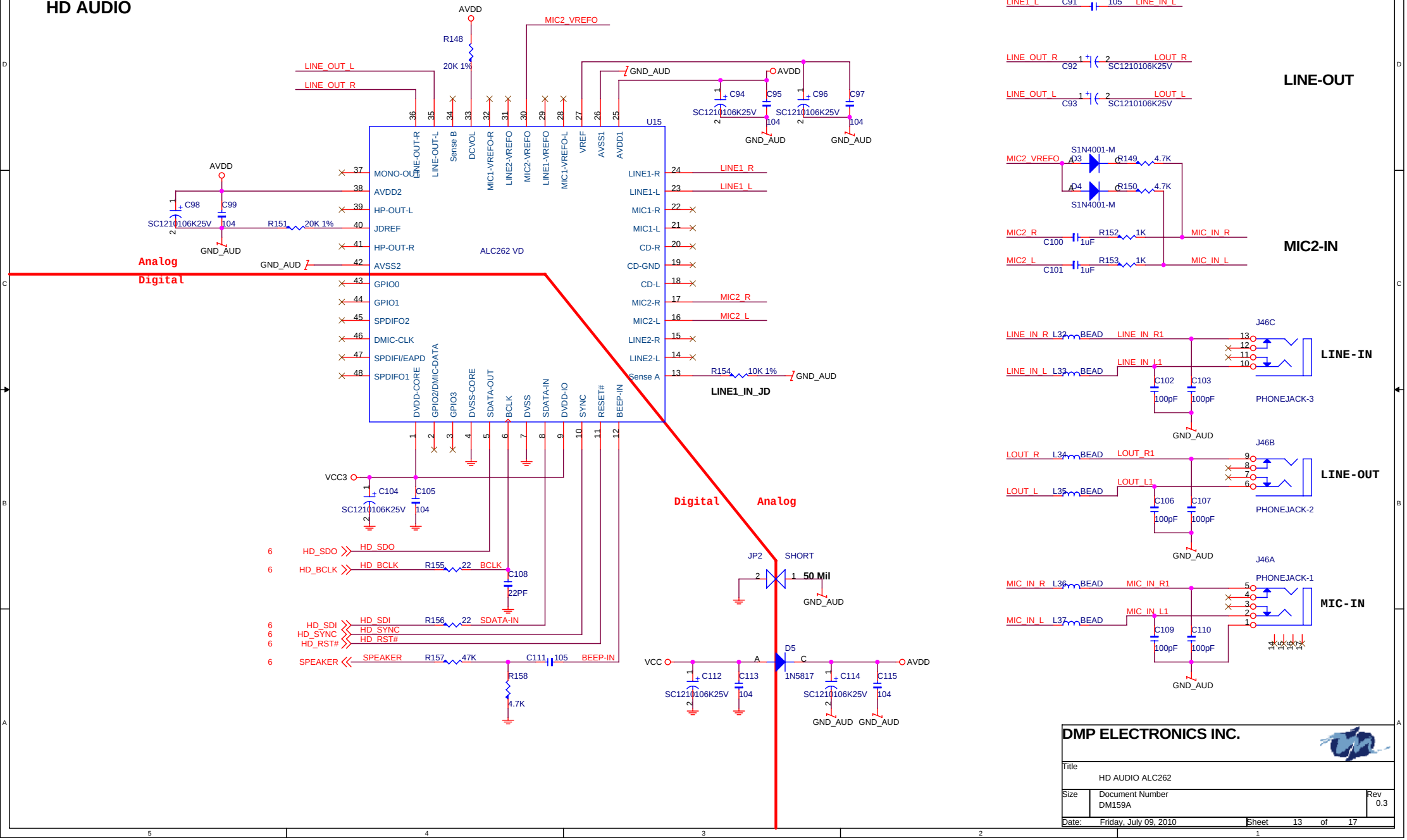
Title	Vortex86SX/DX PRN1/COM3/4	
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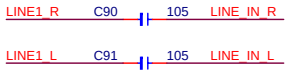
DMP ELECTRONICS INC.

Title	Vortex86SX/DX PCI SLOT/MINI PCI	
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HD AUDIO



LINE1-IN



LINE-OUT

MIC2-IN

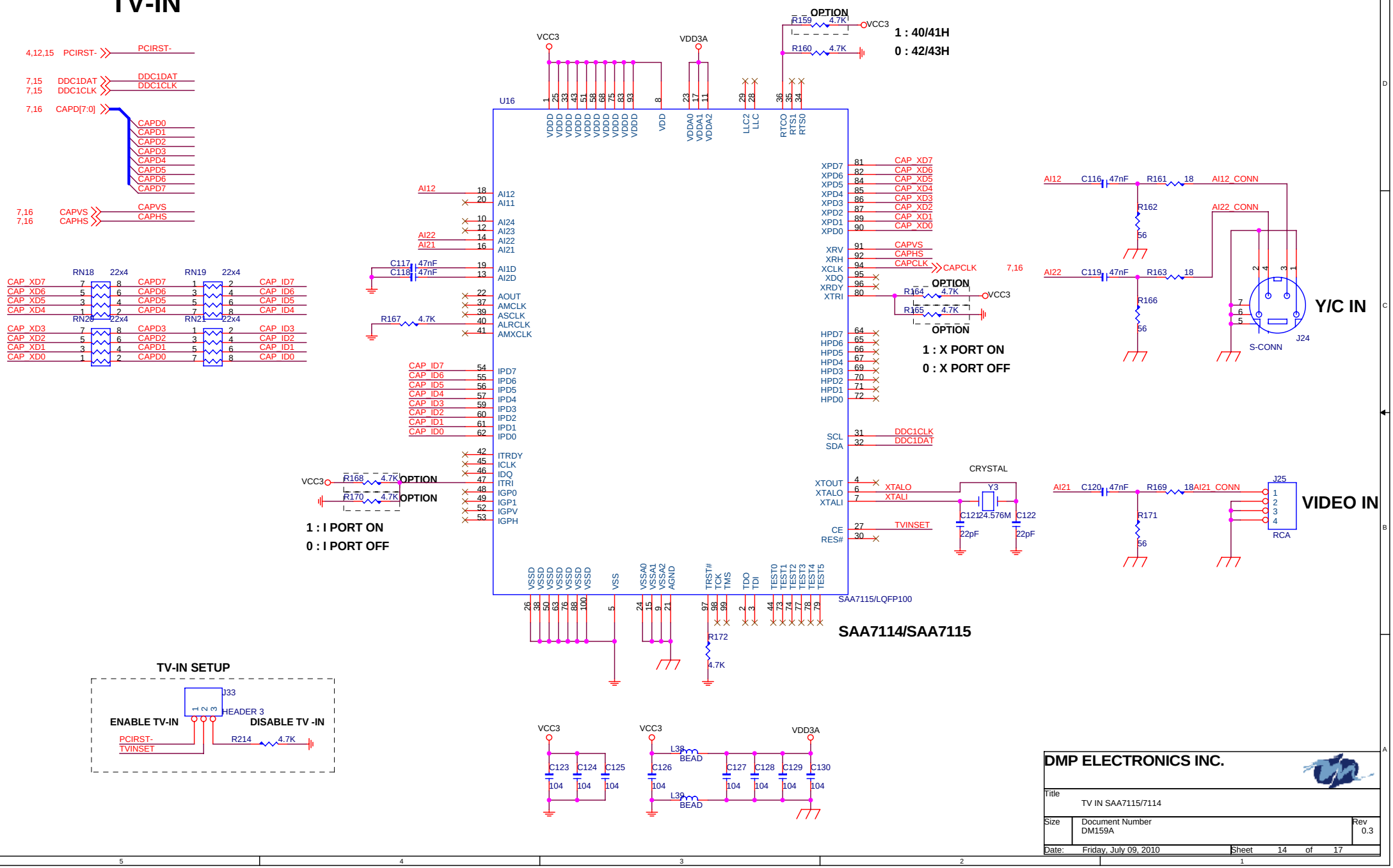
LINE-IN

LINE-OUT

MIC-IN

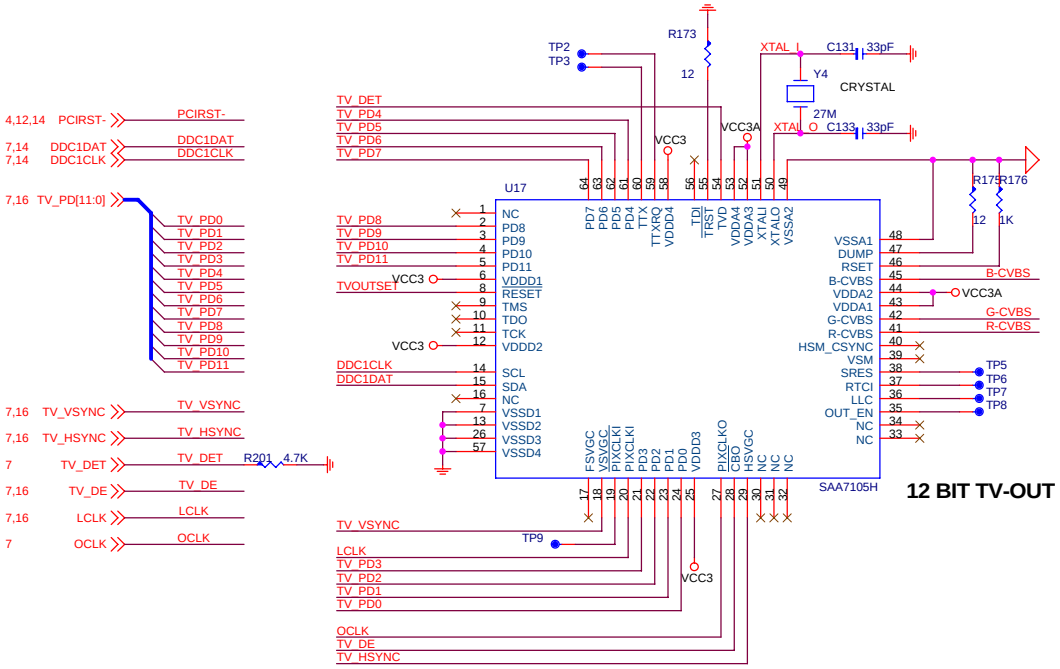
DMP ELECTRONICS INC.		
Title: HD AUDIO ALC262		
Size	Document Number: DM159A	Rev: 0.3
Date: Friday, July 09, 2010	Sheet 13 of 17	

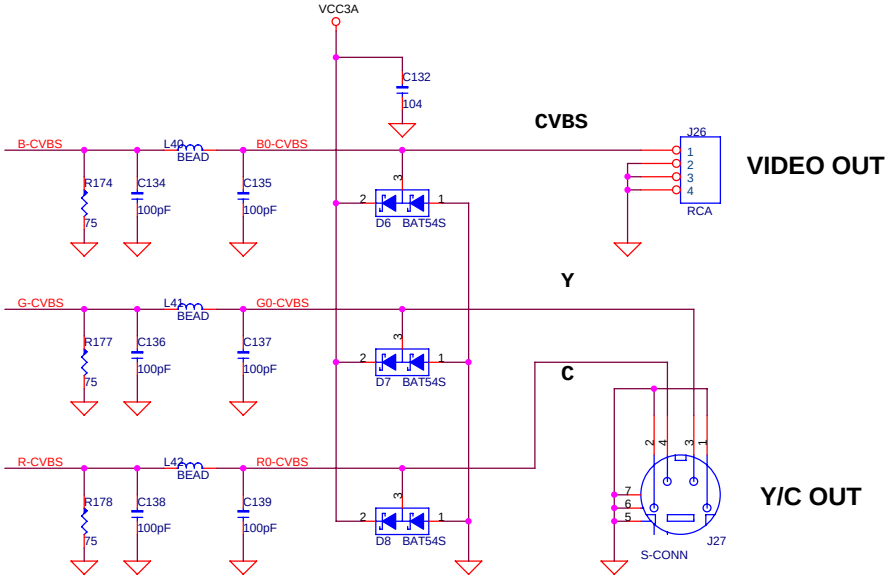
TV-IN

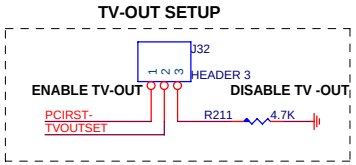


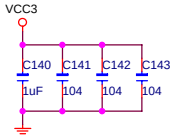
DMP ELECTRONICS INC.		
Title	TV IN SAA7115/7114	
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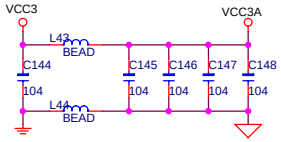
TV-OUT





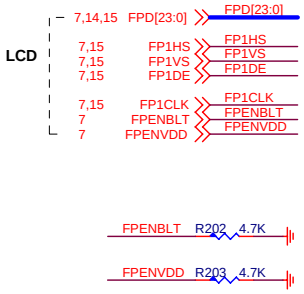


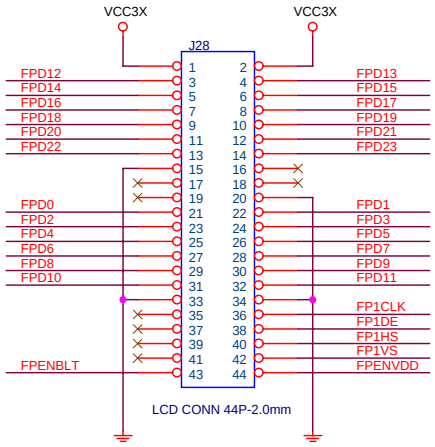




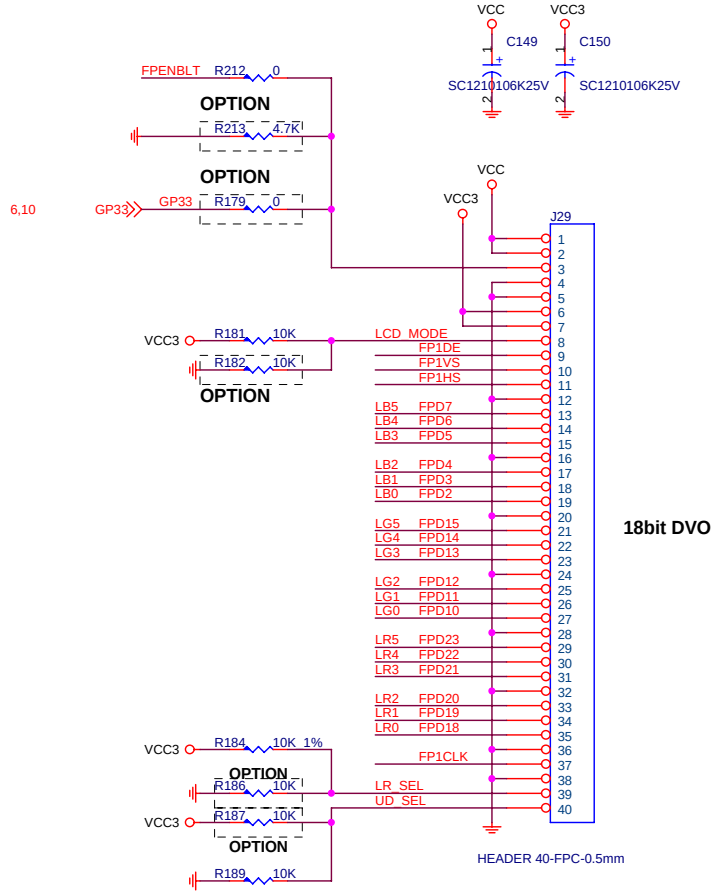
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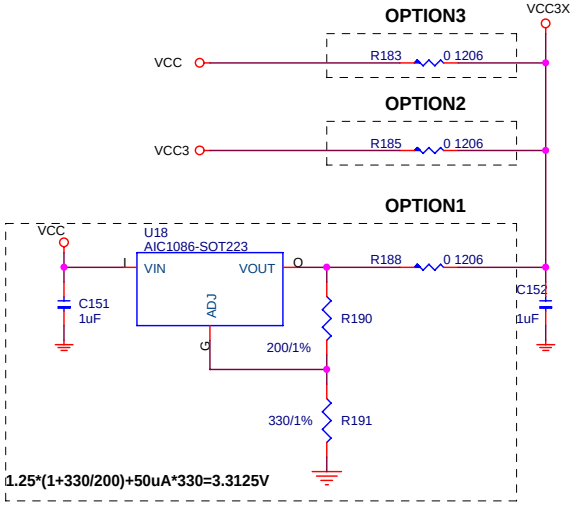
LCD Support 24BIT/18BIT/DSTN





Vortex86MX+ LCD Connector Pin Define





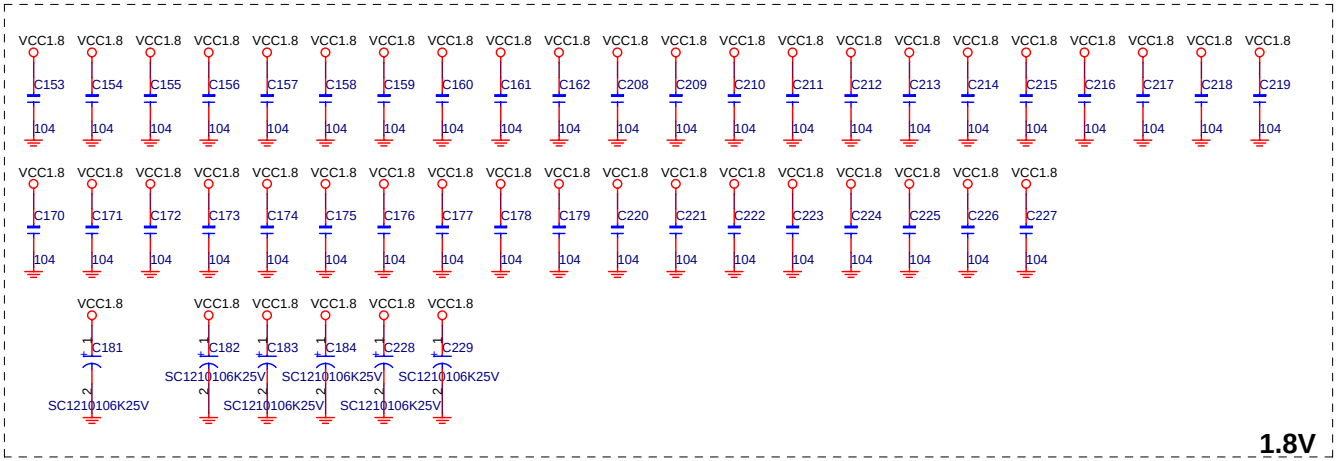
DVO definition table

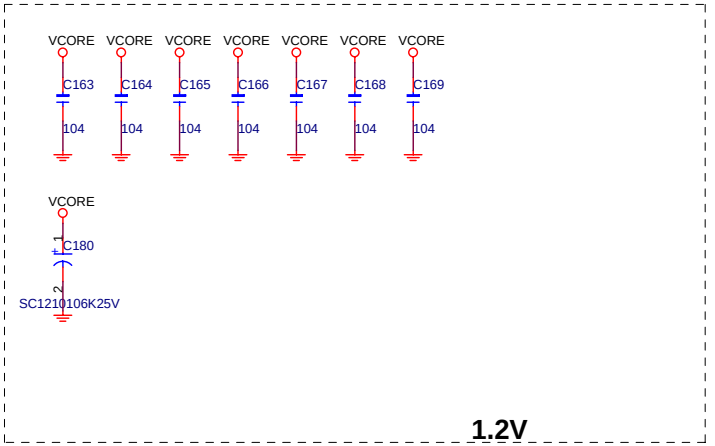
Port / Name / DMP LCD pin	LVDS-24bit SDR	TTL-18bit	LVDS-12bit DDR+CAP	CAP+TV-12bit	DSTN Color	DSTN Mono
FPD23 / LR5	LVD23: R7	D17: R5	CAPCLK	CAPCLK		
FPD22 / LR4	LVD22: R6	D16: R4	VSYNC	VSYNC		
FPD21 / LR3	LVD21: R5	D15: R3	HSYNC	HSYNC		
FPD20 / LR2	LVD20: R4	D14: R2	CAP_FILD	CAP_FILD		
FPD19 / LR1	LVD19: R3	D13: R1	CAPD07	CAPD07	LD7	
FPD18 / LR0	LVD18: R2	D12: R0	CAPD06	CAPD06	LD6	
FPD17	LVD17: R1		CAPD05	CAPD05	LD5	
FPD16	LVD16: R0		CAPD04	CAPD04	LD4	
FPD15 / LG5	LVD15: G7	D11: G5	CAPD03	CAPD03	LD3	LD3
FPD14 / LG4	LVD14: G6	D10: G4	CAPD02	CAPD02	LD2	LD2
FPD13 / LG3	LVD13: G5	D09: G3	CAPD01	CAPD01	LD1	LD1
FPD12 / LG2	LVD12: G4	D08: G2	CAPD00	CAPD00	LD0	LD0
FPD11 / LG1	LVD11: G3	D07:G1	LV1D11:G3R7	TVD11		
FPD10 / LG0	LVD10: G2	D06:G0	LV1D10:G2R6	TVD10		
FPD09	LVD9: G1		LV1D09:G1R5	TVD09		
FPD08	LVD8: G0		LV1D08:G0R4	TVD08		
FPD07 / LB5	LVD7: B7	D05: B5	LV1D07:B7R3	TVD07	HD7	
FPD06 / LB4	LVD6: B6	D04: B4	LV1D06:B6R2	TVD06	HD6	
FPD05 / LB3	LVD5: B5	D03: B3	LV1D05:B5R1	TVD05	HD5	
FPD04 / LB2	LVD4: B4	D02: B2	LV1D04:B4R0	TVD04	HD4	
FPD03 / LB1	LVD3: B3	D01: B1	LV1D03:B3G7	TVD03	HD3	HD3
FPD02 / LB0	LVD2: B2	D00: B0	LV1D02:B2G6	TVD02	HD2	HD2
FPD01	LVD1: B1		LV1D01:B1G5	TVD01	HD1	HD1
FPD00	LVD0: B0		LV1D00:B0G4	TVD00	HD0	HD0
FP1DE	LVDE	DE	LV1DE	TVDE	DISP	
FP1IHS	LVHS	HS	LV1HS	TVHS	LOAD	
FP1VS	LVVS	VS	LV1VS	TVVS	FRM	
FP1CLK	LVCLK	CLK	LV1CLK	TVCLK	CP	
TVCLKIN				TVCLKIN		
FP1DET	DVIDET	DVIDET	DVIDET	TVDET	DVIDET	
ENVEE	ENVEE	ENVEE	ENVEE		ENVEE	
ENPVDD	EPVDD	EPVDD	EPVDD		EPVDD	
ENBLT	ENBL	ENBL	ENBL		ENBL	

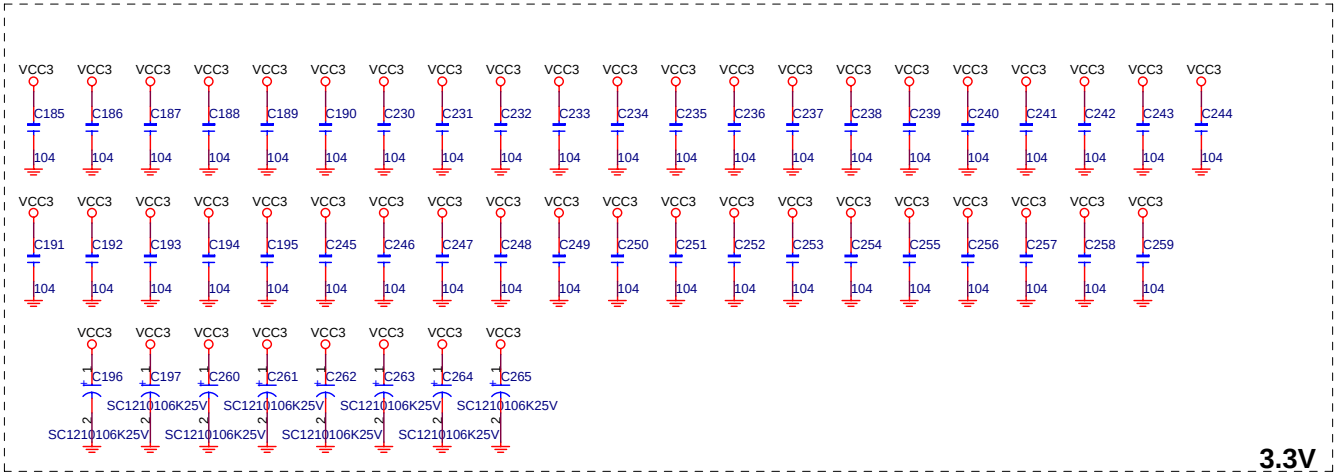
DMP ELECTRONICS INC.

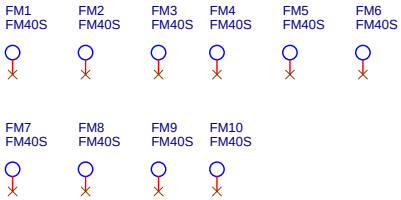
Title	18bit LCD		
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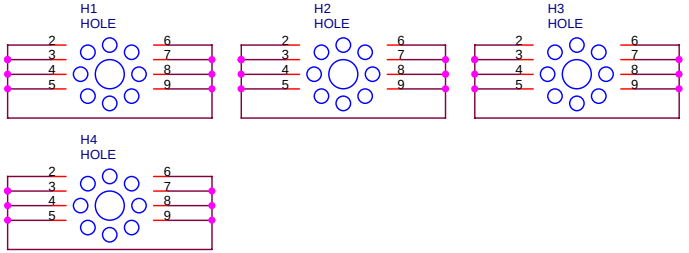
BYPASS

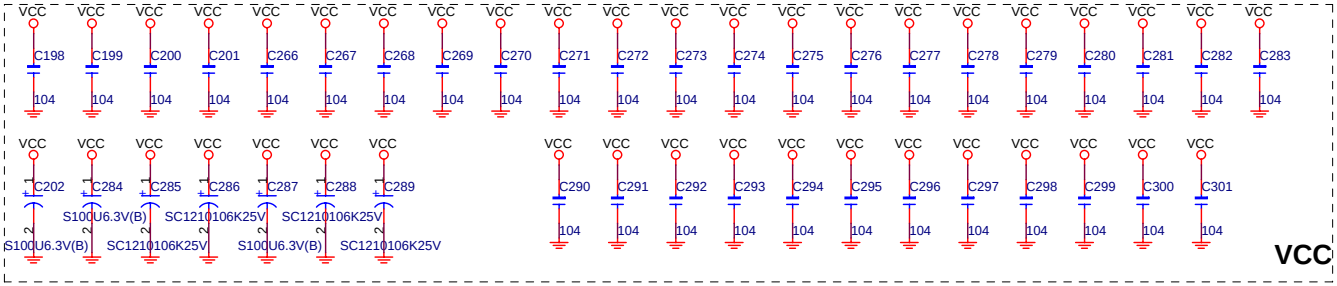












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Title: BYPASS		
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